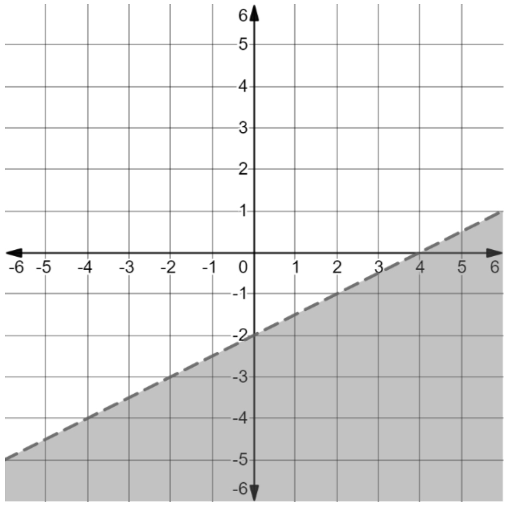
Algebra 1 Name_Answer Key________
Unit 6 Date___________________
Lesson 7 Homework Period__________________

The graph of a *two* variable linear inequality is shown.



1. Write the inequality of the solution set graphed above in slope-intercept form.

   $y < \frac{1}{2}x - 2$

2. Explain how to determine which inequality symbol to use when writing an inequality in slope-intercept form.

   Since the line is dashed, the inequality sign will not be equal to. Since the graph is shaded below the line, select a point in the solution set and use the symbol that make it true.

3. Write the inequality of the solution set graphed above in point-slope form.

   $y + 1 < \frac{1}{2}(x - 2)$ Answers may vary – use any point on the dashed line.

4. Explain how to determine which inequality symbol to use when writing an inequality in point-slope form.

   Since the line is dashed, the inequality sign will not be equal to. Since the graph is shaded below the line, select a point in the solution set shaded region and use the symbol that makes the inequality true.

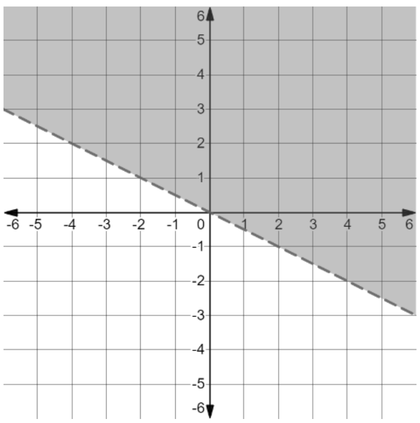
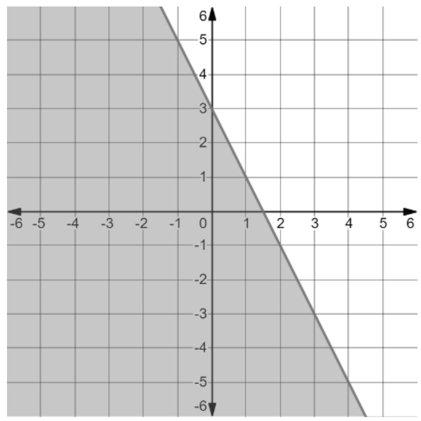
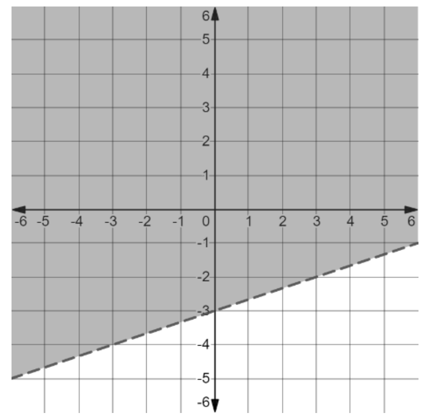
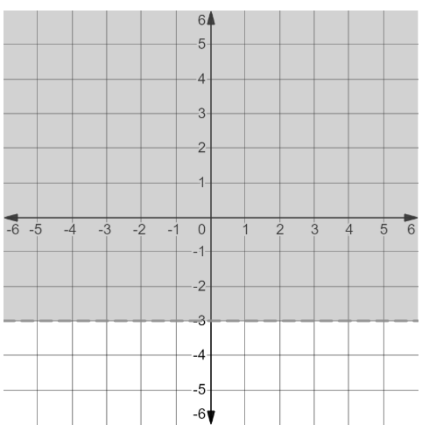
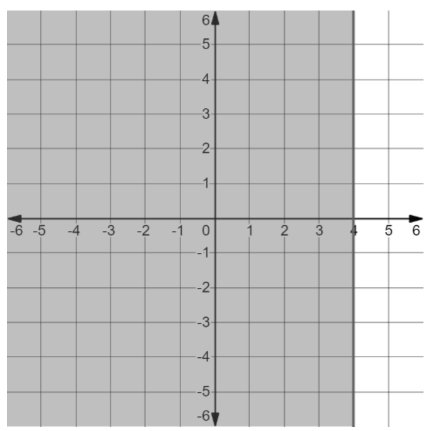
5. Write the inequality of the solution set graphed above in standard form.

   $y < \frac{1}{2}x - 2$ $\rightarrow 2y < x - 2$ $\rightarrow -x + 2y < -2$ $x - 2y > 2$

Five inequalities are given. Complete the table below to match each inequality to the graph of its solution set by writing the inequality on the line provided under each graph.

$x \leq 4$ $\qquad$ $y \leq -2x + 3$ $\qquad$ $-x + 3y > -9$

$y > -3$ $\qquad$ $y + 2 > -\frac{1}{2}(x - 4)$

 6. Inequality: $y + 2 > -\frac{1}{2}(x - 4)$	 7. Inequality: $y \leq -2x + 3$	 8. Inequality: $-x + 3y > -9$
 9. Inequality: $y > -3$	 10. Inequality: $x \leq 4$	